



annual pre-heating power costs

for MTU S4000 G04 gensets

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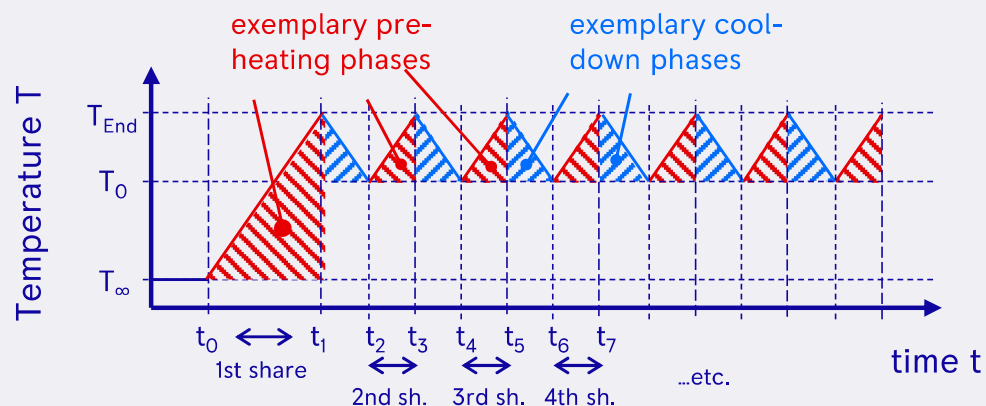
annual power costs of pre-heating unit

for MTU S4000 G04 Gensets



overview

- MTU changes control technique of all (optional) electric pre-heating units for S4000 G04 Gensets
 - in the future pre-heating will start if engine coolant temperature is below $T_0 = 32\text{ °C}$ (former $T_0 = 38\text{ °C}$)
 - coolant will then heat up to $T_{\text{End}} = 43\text{ °C}$ (former $T_{\text{End}} = 49\text{ °C}$)
- **estimation of time shares** in which pre-heating unit is active based on Thermodynamic principles (e.g. 1st law, Newton's law of cooling)



- with new control technique time shares are reduced
→ **less electrical power costs**



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high-temperature (HT) coolant circuit with pre-heating

